

Verifiable research notes on the practical gap between frontier LLMs and near-frontier open models

Executive summary

- **Verified fact — closest public CAISI source.** The closest public, primary CAISI source I could validate for the claim in your screenshot is **NIST CAISI's *Evaluation of DeepSeek AI Models***, organisation-authored by the **Center for AI Standards and Innovation, National Institute of Standards and Technology**, published **30 September 2025**. It is a benchmark report comparing **3 DeepSeek models** with **4 U.S. reference models** across **19 benchmarks**.
- **Verified fact — the public CAISI material retrieved here does not publish an “8 months behind” number.** In the validated CAISI/NIST documents I retrieved, CAISI reports **benchmark-level performance gaps**, not a public “months behind” conversion. The report says DeepSeek lags leading U.S. models, with the largest gaps in **software engineering** and **cyber**, but it does **not** expose a public formula translating that into “6–8 months behind”.
- **Verified fact — an explicit month-lag estimate exists, but from Epoch AI, not the validated CAISI report.** Epoch AI's public analysis states that **Chinese AI models have lagged the U.S. frontier by an average of 7 months since 2023**, with a **4–14 month** range, using the **Epoch Capabilities Index** rather than Elo or CAISI's benchmark table.
- **Verified fact — different evaluation systems measure different things.** CAISI measures task success on selected benchmarks plus security/censorship/adoption; LMSYS/Arena measures **human preference** from pairwise votes using a **Bradley–Terry / Elo-like** framework; Epoch AI uses a **benchmark-stitching latent capability index**; Artificial Analysis uses a **weighted composite index across 10 evaluations**. These are not interchangeable scales.
- **Verified fact — the practical gap is strongly task-dependent.** In CAISI's own data, the gap is **large** on cyber and software-engineering tasks, but **much narrower** on question-answer style knowledge benchmarks. Example: **SWE-bench Verified 66.7 vs 54.8**, but **MMLU-Pro 90.2 vs 89.0** for best U.S. model vs DeepSeek V3.1. CAISI itself says the question-answer benchmarks are “not by a wide margin” and are less indicative of real-world performance than cyber and software-engineering tasks.
- **Verified fact — current open or open-weight models can get surprisingly close on some coding leaderboards.** On Arena's HTML code leaderboard, the top proprietary model in the retrieved snapshot is **Claude Opus 4.7 at 1556**, while the strongest open-weight/open-licensed entries shown include **GLM-5.1 at 1538**, **Kimi K2.6 at 1506**, and **DeepSeek V4 Pro Thinking at 1468**. That is not parity, but neither is it a comic-opera collapse into irrelevance.
- **Verified fact — for retrieval/search-enabled tasks, the retrieved public leaderboard remains proprietary-heavy.** In Arena's Search leaderboard snapshot, the top search-enabled models are proprietary, such as **Claude Opus 4.6 Search at**

1255 and **GPT-5.5 Search at 1235**; the only open model shown in the top 28 snippet is **Diffbot Small XL at 1024**. That suggests the gap matters more when search/retrieval orchestration is central.

- **Interpretation — the headline claim is directionally plausible, but source attribution is shaky.** “Near-frontier/open models are ~6–8 months behind” is **well aligned with Epoch AI’s explicit public analysis**, but **not directly substantiated by the public CAISI documents retrieved here**. The second half of the claim — “this may not matter in practice” — is partially supported by the narrower gaps on Q&A-style tasks and by coding leaderboards where strong open-weight models are close on some slices, but it does **not** hold uniformly across cyber, retrieval/search, or agentic engineering workloads.

CAISI study validation

Source identity

Item	Extracted note
Title	<i>Evaluation of DeepSeek AI Models</i>
Authors / organisation	Center for AI Standards and Innovation, National Institute of Standards and Technology
Publication date	30 September 2025
Original source	NIST-hosted PDF and NIST news release

- **Verified fact — source metadata.** *Evaluation of DeepSeek AI Models* is a NIST-hosted CAISI report; the PDF cover identifies **Center for AI Standards and Innovation** and **National Institute of Standards and Technology**, and the NIST news page dates the release to **30 September 2025**.
- **Verified fact — related public CAISI follow-up.** A later NIST update, **CAISI Evaluation of Kimi K2 Thinking**, published **12 December 2025**, states that Kimi K2 Thinking was the **most capable PRC-based model at the time of release**, while still lagging leading U.S. models.
- **Verified fact — unresolved attribution issue.** The chart in your screenshot appears to use an **Elo-style “overall capability” trajectory**, but the validated public CAISI PDF uses **benchmark success percentages**, security results, censorship results, and adoption metrics; I did **not** find a public CAISI report in this pass that reproduces the screenshot’s specific Elo chart and labels.

What CAISI actually measures

- **Verified fact — CAISI capability metric is benchmark success, not Elo.** The report evaluates models across four capability domains — **cyber, software engineering, science and knowledge, mathematical reasoning** — and reports **accuracy / percentage of tasks solved** on the included benchmarks.
- **Verified fact — benchmark set.** The report uses **19 benchmarks** in total, including public and CAISI/private benchmarks. The capability table includes **CVE-Bench, Cybench, CTF-Archive, SWE-bench Verified, Breakpoint, MMLU-Pro, MMMLU, GPQA, HealthBench, Humanity’s Last Exam, SMT 2025, OTIS-AIME 2025, and PUMaC 2024**.

- **Verified fact — agentic evaluations were normalised.** For cyber and software tasks, CAISI used a standardised **ReAct-style agent loop**, a common Linux/Docker environment, and a **fixed 500,000 weighted-token budget** per agentic task.
- **Verified fact — uncertainty modelling.** CAISI estimates benchmark uncertainty using **mixed-effects logistic regression**, treating model as a fixed effect and task as a random effect; it reports error bars as **estimated standard errors**.

What “months behind” does and does not mean in CAISI

- **Verified fact — public CAISI wording.** CAISI says DeepSeek performance **“lags behind”** leading U.S. models; the best U.S. model outperforms DeepSeek V3.1 on almost every benchmark, with the largest gaps in **software engineering** and **cyber**.
- **Verified fact — public CAISI does not quantify lag in months.** The retrieved public CAISI materials do **not** state “8 months behind” and do **not** publish a method for converting benchmark gaps into elapsed calendar time.
- **Interpretation — likely source confusion.** The phrase “about 8 months behind” is more plausibly a **secondary interpretation** or a borrowing from other benchmark frameworks such as **Epoch AI’s 7-month estimate**, rather than a direct quotation from the validated NIST CAISI report.

Method assumptions and limitations

- **Verified fact — server path matters.** CAISI evaluated proprietary U.S. models **through APIs**, but downloaded DeepSeek weights from **Hugging Face** and ran them on CAISI cloud servers. The report explicitly says evaluations on **DeepSeek APIs or third-party hosts may yield different results**, especially for security and censorship.
- **Verified fact — the report is partial and preliminary.** CAISI’s disclaimer says the findings are **preliminary**, limited to **specific domains**, **model versions**, and **evaluation methods**, and may not generalise to other tasks. It also notes benchmark limitations, including relevance to practical tasks and possible overlap with training data.
- **Verified fact — cost comparison caveat.** CAISI compares **end-to-end expense**, not just token price, and explicitly notes that DeepSeek’s lower pricing came with **higher latency** and a **smaller maximum prompt size** (**64k** vs “many U.S. cloud providers” at **128k**), which may make its analysis conservative in DeepSeek’s favour.
- **Verified fact — the report does not address all strategic questions.** CAISI says it does **not** investigate how models were developed, the cost of development, or whether they were trained on distilled data.

Alternative analyses

Comparison table

Source	Primary metric	Public conclusion relevant to your claim	Raw data / methodology availability
LMSYS / Arena	Human-preference Bradley-Terry / Elo-like score from anonymous pairwise votes	Useful for “how good does it feel in practice?” on open-ended tasks; not a month-lag estimate	Method paper + public leaderboard dataset

Source	Primary metric	Public conclusion relevant to your claim	Raw data / methodology availability
Epoch AI	Epoch Capabilities Index from stitched benchmark scores	Chinese models lagged U.S. frontier by 7 months on average since 2023	Public methodology page + public code repo
Artificial Analysis	Weighted Intelligence Index across 10 evals	Gap to leading U.S. proprietary models is narrower than in earlier periods; Chinese open-weight models have periodically led the open-weight frontier	Public methodology + live leaderboard/evaluation pages

LMSYS / Arena

- **Verified fact — methodology.** The Chatbot Arena paper describes an **open platform for evaluating LLMs by human preference** via **pairwise comparisons** collected from users; Arena’s current FAQ says rankings are computed with the **Bradley–Terry rating system**, similar to Elo.
- **Verified fact — what it measures.** Arena is strongest as a measure of **user-perceived quality on open-ended prompts**, rather than “objective capability” in the CAISI/Epoch sense. Arena itself now refers to the score as **Arena score** rather than “Elo”, because it is reporting Bradley–Terry coefficients on an Elo-like scale.
- **Verified fact — style confounds are real.** Arena has published a technical note showing that rankings can be affected by response **length** and **markdown/style**, and that it uses **style control** in Bradley–Terry regression to reduce that confounding.
- **Verified fact — current evidence from one task slice.** On Arena’s HTML code leaderboard snapshot, proprietary leaders still occupy the top positions, but open/open-weight models are close: **Claude Opus 4.7 1556**, **GLM-5.1 1538**, **Kimi K2.6 1506**, **DeepSeek V4 Pro Thinking 1468**.
- **Agreement with CAISI.** Frontier proprietary models still dominate the overall top of the leaderboard.
- **Disagreement or non-overlap with CAISI.** Arena is not a security/censorship benchmark and does not provide a country-level month-lag estimate. It may show smaller practical differences on user-facing tasks than CAISI’s cyber/software-engineering tables.

Epoch AI

- **Verified fact — explicit month-lag result.** Epoch AI’s analysis states that since 2023, Chinese AI models have trailed the U.S. frontier by an **average of 7 months**, with a **minimum of 4** and a **maximum of 14** months.
- **Verified fact — methodology.** Epoch computes lag using the **Epoch Capabilities Index**, which stitches benchmarks into a single capability scale. For the country-gap analysis, it tracks the **best model from each country on each day**, then asks how long since the leading U.S. model last had an ECI score as low as the best Chinese model’s score on that day, treating scores within **1 ECI point** as equivalent.
- **Verified fact — what ECI is.** ECI uses **Item Response Theory-style** modelling with benchmark **difficulty** and **slope** parameters; the current methodology page says it draws on **37 distinct benchmarks** and is fit by **non-linear least squares with ridge regularisation**.

- **Verified fact — benchmark breadth.** Epoch’s current methodology page lists a broad mix of internal evaluations, independent leaderboards, and developer-reported benchmarks, including **GPQA Diamond, FrontierMath, SimpleQA Verified, Aider polyglot, SWE-bench (Bash Only), Terminal-Bench, Cybench, MMLU, OSWorld, and others.**
- **Agreement with the headline claim.** This is the cleanest public source in the retrieved material for the “~6–8 months behind” statement.
- **Disagreement with CAISI.** Epoch’s lag is a **composite statistical translation across many benchmarks**, whereas CAISI publishes **task-specific gaps** and adds national-security evaluations such as hijacking, jailbreaking, and censorship. The two sources are complementary, not directly comparable.

Artificial Analysis

- **Verified fact — methodology.** Artificial Analysis’ current **Intelligence Index v4.0.4** is a weighted average over **10 evaluations**: **GDPval-AA, τ^2 -Bench Telecom, Terminal-Bench Hard, SciCode, AA-LCR, AA-Omniscience, IFBench, Humanity’s Last Exam, GPQA Diamond, and CritPt**, grouped into four equally weighted categories.
- **Verified fact — current live snapshot.** In the retrieved live leaderboard snapshot, **GPT-5.5 (xhigh)** scores **60**, while leading Chinese/open-weight or near-frontier models include **Kimi K2.6 at 54, DeepSeek V4 Pro (Max) at 52, Qwen3.6 Max Preview at 52**, and **gpt-oss-120B (high) at 33**.
- **Verified fact — prior China report.** Artificial Analysis’ Q2 2025 China report says the U.S.–China frontier gap had become **“as narrow now as it has ever been”**, and that the Chinese **open-weight frontier surpassed the U.S. open-weight frontier** in **November 2024**.
- **Verified fact — explicit caveat.** Artificial Analysis’ China report says that **some results were estimated based on company claims and comparable results**, rather than all being independently benchmarked at the moment of publication.
- **Agreement with the headline claim.** Artificial Analysis supports the view that strong Chinese/open-weight models are now **near-frontier on several dimensions**, especially in the open-weight segment.
- **Disagreement with CAISI in emphasis.** CAISI stresses **security, censorship, and cyber/software gaps**; Artificial Analysis places more weight on **general capability composites, coding, agentic workflows, and price/performance**.

Capability gaps by task type

Quantitative examples

Task category	Metric	Frontier / best-U.S. example	Near-frontier / open example	Extracted gap	Source
Reasoning and coding	SWE-bench Verified	66.7	54.8(DeepSeek V3.1)	11.9 pts	CAISI
Reasoning and coding	Breakpoint	98.0	78.5(DeepSeek V3.1)	19.5 pts	CAISI
Math reasoning	OTIS-AIME 2025	91.9	77.6(DeepSeek V3.1)	14.3 pts	CAISI
General knowledge / chat-adjacent	MMLU-Pro	90.2	89.0(DeepSeek V3.1)	1.2 pts	CAISI

Task category	Metric	Frontier / best-U.S. example	Near-frontier / open example	Extracted gap	Source
General knowledge / chat-adjacent	GPQA	86.9	79.3(DeepSeek V3.1)	7.6 pts	CAISI
General knowledge / hard synthesis	Humanity's Last Exam	26.6	13.0(DeepSeek V3.1)	13.6 pts	CAISI
Search / retrieval-enabled	Arena Search score	1255 (Claude Opus 4.6 Search)	1024 (Diffbot Small XL, open)	231 Arena-score pts	Arena
Coding in a user-facing arena	Arena HTML code score	1556 (Claude Opus 4.7)	1538 (GLM-5.1, MIT)	18 Arena-score pts	Arena

Reasoning and coding

- **Verified fact — CAISI shows large practical gaps on software engineering.** In CAISI's table, the best U.S. model scores **66.7** on **SWE-bench Verified** versus **54.8** for DeepSeek V3.1, and **98.0** vs **78.5** on **Breakpoint**. CAISI's summary says the gap is largest for **software engineering** and **cyber**.
- **Verified fact — CAISI shows moderate-to-large gaps on maths.** On CAISI's maths benchmarks, the best U.S. model scores **91.9 vs 77.6** on **OTIS-AIME 2025**, **85.9 vs 77.7** on **PUMaC 2024**, and **91.8 vs 86.2** on **SMT 2025**.
- **Verified fact — GPT-5 also improved materially over OpenAI's prior frontier.** OpenAI's GPT-5-for-developers launch states **GPT-5 scores 74.9% on SWE-bench Verified versus o3 at 69.1%**, and **88% on Aider polyglot**.
- **Verified fact — but open/open-weight models can be close on some live coding arenas.** On Arena's HTML code leaderboard snapshot, **GLM-5.1 (MIT)** is only **18 Arena-score points** behind **Claude Opus 4.7**, while **Kimi K2.6** and **DeepSeek V4 Pro Thinking** are also in the upper tier.
- **Interpretation.** For coding, the gap is real but not uniform: frontier advantage is clearer on curated engineering benchmarks and some agentic tasks than on all user-facing coding arenas.

General chat and summarisation

- **Verified fact — CAISI's narrowest gaps are on Q&A-style knowledge benchmarks.** The best U.S. model scores **90.2 vs 89.0** on **MMLU-Pro**, **87.7 vs 82.2** on **MMMLU**, and **63.0 vs 52.5** on **HealthBench** relative to DeepSeek V3.1.
- **Verified fact — CAISI's own interpretation is that these gaps are narrower.** CAISI says U.S. and DeepSeek models achieve **similar performance** on question-answer-style science and knowledge benchmarks, with frontier models only **slightly more performant**, and "not by much".
- **Verified fact — human-preference frameworks are relevant here.** LMSYS/Arena's entire methodology is based on **human preference in pairwise chat comparisons**, making it a useful complement for general-chat use cases that fixed multiple-choice benchmarks often miss.
- **Interpretation.** This is the strongest support in the retrieved material for the claim that "the gap may not matter in practice" — if the application is mostly drafting, summarising, or knowledge synthesis rather than hard agentic coding or cyber tasks.

RAG and retrieval-based systems

- **Verified fact — Search Arena is a direct benchmark for search-enabled assistants.** Arena’s Search leaderboard evaluates search-enabled AI systems and in the retrieved snapshot is led by proprietary models such as **Claude Opus 4.6 Search (1255)**, **GPT-5.5 Search (1235)**, and **Claude Sonnet 4.6 Search (1221)**.
- **Verified fact — open presence in that snapshot is thin.** The retrieved top-28 Search Arena snippet includes **Diffbot Small XL (Apache 2.0)** at **1024**, well below the leading proprietary search-enabled systems.
- **Interpretation.** If your “RAG system” is really a **search-and-browse assistant with tool use**, the grade-school slogan “all good models are basically the same” looks rather less convincing. If it is classic enterprise RAG over a stable private corpus with light synthesis, the retrieved packet here is less conclusive.
- **Open gap in this research pass.** I did **not** retrieve a high-confidence, primary-source benchmark in this pass that cleanly compares frontier vs open models on **enterprise private-corpus RAG** with identical retrieval stacks. That should be treated as **unspecified / still needed**.

Structured tasks

- **Verified fact — Artificial Analysis includes a structured instruction-following benchmark.** The Intelligence Index includes **IFBench**, with **294 tasks**, **5 repeats**, **rule-driven assessment**, and **pass@1** scoring.
- **Verified fact — but per-model IFBench scores were not retrieved in this pass.** The methodology is available, but I did not retrieve the corresponding per-model score table here. That detail should be marked **unspecified in this pass** rather than guessed.
- **Best proxy from retrieved evidence.** CAISI’s small gap on **MMLU-Pro** and moderate gap on **MMMLU** suggest that structured knowledge/classification-like tasks may compress the frontier advantage, but **HLE** shows that the gap widens again as task difficulty rises.

Cost and operational notes

What is directly verified in the retrieved sources

Type	Extracted fact	Status
End-to-end cost	GPT-5-mini was 35% less costly on average than DeepSeek V3.1 at similar performance across CAISI’s 13 capability benchmarks	Verified
Latency / context trade-off	CAISI says DeepSeek’s lower price came with higher latency and 64k context , vs many U.S. providers at 128k	Verified
Current blended live prices	AA snapshot shows GPT-5.5 (xhigh) \$11.25 blended / 1M tokens , DeepSeek V4 Pro (Max) \$2.17 , Kimi K2.6 \$1.71 , gpt-oss-120B (high) \$0.26	Verified, third-party live aggregator, not official pricing page
Current coding price/performance examples	Arena HTML shows Opus 4.7 \$5 / \$25 , GLM-5.1 \$1.40 / \$4.40 , DeepSeek V4 Pro Thinking \$0.43 / \$0.87 , MiniMax-M2.7 \$0.30 / \$1.20	Verified, leaderboard-captured public prices, not official pricing page
• Verified fact — measured user cost can reverse a “cheap open model” story. CAISI’s cost analysis explicitly uses end-to-end expense per task, not list token		

price, and finds **GPT-5-mini** cheaper than **DeepSeek V3.1** on **11 of 13** capability benchmarks, averaging **35% lower expense** despite similar overall performance class.

- **Verified fact — latency and context are part of the economic story.** CAISI says DeepSeek makes user-experience trade-offs to lower price, including **higher latency** and a **smaller prompt limit**, and warns that its end-to-end expense estimate may therefore **underestimate** DeepSeek's effective cost disadvantage.
- **Verified fact — live market prices still show large price dispersion.** Artificial Analysis' live snapshot shows a very wide spread between frontier proprietary and open/open-weight models in blended per-token cost, from **\$11.25** for **GPT-5.5 (xhigh)** down to **\$0.26** for **gpt-oss-120B (high)** and low single-digit dollars for several strong Chinese/open-weight models.
- **Interpretation.** For a blog post, do not treat "open-weight = cheaper in practice" as a universal law. The strongest quantitative evidence retrieved here supports a subtler statement: **token price can be much lower**, but **end-to-end task cost** may still favour a smaller or better-optimised frontier API on specific workloads.

Operational complexity

- **Verified fact — evaluating agentic systems is operationally heavy.** CAISI's appendix shows that agentic evaluation required a **Docker/Linux environment**, **persistent shells**, **Python execution**, **file tools**, model-specific tool-calling optimisation, and a standardised token budget.
- **Verified fact — fair evaluation itself needs engineering controls.** Arena publishes dedicated work on **style control** in its Bradley–Terry regression, showing that response formatting and verbosity can bias outcomes if not corrected.
- **Verified fact — Artificial Analysis' agentic benchmarks require tool and sandbox orchestration.** For GDPval-AA, Artificial Analysis runs models inside an **E2B sandbox** and exposes tools such as **web fetch**, **web search**, **shell execution**, and **finish**, then grades outputs pairwise with Elo-style scoring.
- **Interpretation.** In real deployments, the operational burden often shifts from "which model is smartest?" to "which stack is easiest to run, monitor, secure, and pay for?" It is annoyingly mundane, which is exactly why it matters.

What remains incomplete

- **Open gap — official pricing pages.** I did **not** retrieve the official pricing pages for **Anthropic, Together, Groq, AWS, GCP, or Azure GPU rental** in this pass, so those should be marked **unspecified / still needed** rather than inferred. The cost notes above therefore rely on **CAISI's measured cost comparison** and **live third-party leaderboard snapshots**, not a complete official-pricing audit.
- **Open gap — self-hosting economics.** I did **not** retrieve high-confidence source material in this pass for **A100/H100 rental prices, purchase prices, throughput assumptions, or 1M / 100M request self-hosting estimates**. Those items remain **unfilled**.

Strategic implications and open questions

Grounded examples

- **Verified fact — frontier model usage in knowledge work.** OpenAI's GPT-5.5 launch article says **more than 85% of OpenAI uses Codex every week**. It also reports internal deployments where the Comms team analysed **six months of speaking-request data**, the Finance team reviewed **24,771 K-1 tax forms across 71,637 pages**, finishing **two weeks faster** than the prior year, and a go-to-market workflow saved **5–10 hours per week** on weekly reporting.
- **Verified fact — frontier buyers cite autonomy and persistence.** In the same OpenAI source, early testers such as Cursor and MagicPath attribute GPT-5.5's value to stronger **reasoning**, **tool use**, **persistence on long-running tasks**, and the ability to carry changes through larger systems with less correction required.
- **Verified fact — open-weight models have strong ecosystem adoption evidence.** CAISI reports that DeepSeek model downloads on model-sharing platforms rose by **nearly 1000% since January 2025**; DeepSeek API usage on OpenRouter increased by **over 5,900%** from January to September 2025; and models derived from PRC base models uploaded to Hugging Face now outnumber those derived from U.S. developers, driven mainly by **Alibaba** and **DeepSeek**.
- **Verified fact — open-weight ecosystem competition is not theoretical.** CAISI's benchmark summary says the U.S. open-weight model it evaluated, **gpt-oss**, outperformed the best DeepSeek models on **most agentic tasks**, while the best DeepSeek models slightly outperformed gpt-oss on **science/knowledge** and **mathematics**.
- **Verified fact — Artificial Analysis also attributes Chinese adoption partly to openness.** Its China report says DeepSeek and Alibaba have both embraced **open-weight strategies**, supporting widespread domestic and international adoption; it also states that China's open-weight frontier surpassed the U.S. open-weight frontier in late 2024.

Interpretable takeaways for your later blog

- **Interpretation — when the gap probably matters.** The retrieved evidence says the gap matters more for **agentic coding**, **cyber/security-sensitive tasks**, **search/retrieval-heavy assistants**, and very difficult reasoning benchmarks such as **HLE**.
- **Interpretation — when the gap may matter less.** The gap is smaller for **question-answer style knowledge tasks**, some **general drafting/summarisation-like workloads**, and some **UI/web coding arenas** where well-tuned open/open-weight models remain very competitive.
- **Interpretation — the strongest careful formulation.** A rigorous claim supported by the retrieved material would be: **"Near-frontier and open-weight models often trail frontier proprietary models on broad capability composites and agentic benchmarks; however, for many knowledge-heavy and some coding workloads, the observed gap is substantially smaller than headline 'frontier gap' narratives suggest."**

Open questions and limitations

- **Unresolved source-validation question.** I did **not** validate a public CAISI/NIST document containing the screenshot's **Elo-style capability chart** or the exact phrase **"8 months behind"**. The validated public CAISI materials retrieved here are the **DeepSeek evaluation** and the **Kimi K2 Thinking update**.
- **Incomplete cost audit.** Official pricing pages for **Anthropic, Together, Groq, AWS, GCP, Azure**, plus **self-hosting rental/purchase/throughput** evidence, were **not retrieved in this pass**. Use the current cost section as a **partial evidence base**, not a complete pricing chapter.
- **Incomplete company case studies for open models.** I retrieved strong **ecosystem adoption evidence** for open/open-weight models, but I did **not** retrieve enough high-confidence, company-specific case studies in this pass to satisfy a thorough "Which companies chose open models, and why?" section. Mark that as **still needed**.
- **Metric non-equivalence warning.** Do **not** compare CAISI percentages, Arena scores, Epoch ECI values, and Artificial Analysis index points as if they were commensurate. They answer different questions, with different data-generating processes. That way lies chart crime, and worse, a confident paragraph on LinkedIn.



Bibliography (Structured)

1) Core "Frontier vs Open Models" Analysis

Primary (claimed source to validate)

- Center for AI Standards and Innovation (CAISI)
 - *[Exact report name to confirm via research step]*
 - Focus: Model capability gap vs time ("months behind" framing)

Alternative / Supporting Analyses

- LMSYS Chatbot Arena
 - <https://chat.lmsys.org/?leaderboard>
 - Method: Human preference rankings (pairwise voting)
- Epoch AI
 - <https://epochai.org/>
 - Focus: Compute trends, scaling laws, capability forecasting
- Artificial Analysis
 - <https://artificialanalysis.ai/>
 - Focus: Model benchmarks, pricing, latency comparisons

2) Benchmarking and Capability Measurement

- HELM Benchmark (Stanford CRFM)

- <https://crfm.stanford.edu/helm/latest/>
 - MMLU Benchmark
 - <https://arxiv.org/abs/2009.03300>
 - BIG-bench
 - <https://github.com/google/BIG-bench>
 - Open LLM Leaderboards (Hugging Face)
 - https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard
-

3) Cost and Infrastructure Sources

API Pricing (Frontier Models)

- OpenAI Pricing
 - <https://openai.com/pricing>
- Anthropic Pricing
 - <https://www.anthropic.com/pricing>

Open-weight / Inference Providers

- Together AI
 - <https://www.together.ai/pricing>
- Groq
 - <https://groq.com/>
- Fireworks AI
 - <https://fireworks.ai/>

Cloud GPU Pricing

- AWS EC2 GPU Instances
 - <https://aws.amazon.com/ec2/instance-types/>
- Google Cloud GPUs
 - <https://cloud.google.com/compute/gpus-pricing>
- Azure GPU Pricing
 - <https://azure.microsoft.com/en-us/pricing/details/virtual-machines/>

Hardware (On-premise estimates)

- NVIDIA H100 / A100 specs
 - <https://www.nvidia.com/en-us/data-center/>
-

4) Operational and Engineering Trade-offs

- Google: “Rules of Machine Learning”
 - <https://developers.google.com/machine-learning/guides/rules-of-ml>

- Meta Engineering Blog (LLM infra)
 - OpenAI System Cards (deployment insights)
-

5) Industry Case Studies (Ranking, ML systems)

Airbnb (core set you already analysed)

- *Applying Deep Learning to Airbnb Search (2018)*
→ Scaling simple architectures + data beats complexity
 - *Improving Deep Learning for Airbnb Search (2020)*
→ Post-launch iteration, ICE plots, two-tower models, bias fixes
 - *Real-time Personalisation with Embeddings (Part 1)*
→ Skip-gram embeddings from click sequences
 - *Real-time Personalisation with Embeddings (Part 2)*
→ User modelling, sparsity handling, embedding features in ranking
 - *Airbnb Location Model (2013)*
→ Feature engineering > brute-force modelling
 - *Managing Diversity in Airbnb Search*
→ When relevance collapses diversity
-

Booking.com

- *Beyond Algorithms: Ranking at Scale (2020)*
→ Bias correction, experimentation, large-scale ranking systems
-

6) Key Themes You Can Cite Explicitly

From these sources, you can anchor your blog around:

- **Capability gap exists but is contextual**
- **Cost scales non-linearly with frontier models**
- **Engineering complexity dominates model choice**
- **Simple models + good features still win in production**
- **Iteration and diagnostics matter more than architecture**

For example, Airbnb explicitly shows that:

- Scaling data + simple architectures beat complex hybrids
- Model improvements often come from **problem reframing, not model upgrades**

Which, if you are being slightly cheeky, undermines half of Twitter.

7) How to Use This in Your Blog

A clean way to integrate this:

- **Section 1:** “The 6–8 month gap” → cite CAISI + LMSYS
- **Section 2:** “What capability actually means” → benchmarks
- **Section 3:** “The cost reality” → pricing + GPUs
- **Section 4:** “Production reality” → Airbnb + Booking
- **Section 5:** “So what?” → your thesis